



AI LICENSE PLATE RECOGNITION

FOR GATED COMMUNITY ACCESS



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THE PROBLEM I AM SOLVING

- The daily frustrations of accessing gated apartment complexes. -

✗ CURRENT CHALLENGES

NEED TO TYPE A CODE



Slow, inconvenient, and wastes time every day.

LOSING REMOTE CONTROL



Easy to lose, forget, or drop without realizing.

BATTERY DIES AT THE WORST TIME



During storms you have to open the window and type the code.



✓ MY SOLUTION

AI LICENSE PLATE RECOGNITION FOR GATED COMMUNITY ACCESS

The system automatically detects the license plate, verifies it against the database, and opens the gate for authorized residents.



IMPACT: A BETTER, SAFER, AND SMARTER EXPERIENCE FOR MILLIONS



Millions of people live in gated communities and face these same frustrations every day.



Saves time and makes entering the community fast, easy, and stress-free.



Improves security by allowing only authorized vehicles to enter the community.



Reduces traffic, idling, and unnecessary stops, helping the environment and our community.



A SMALL CHANGE
THAT MAKES
A BIG DIFFERENCE
FOR MILLIONS.

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Smart • Secure • Automated



SECURE

Only authorized vehicles are allowed.



FAST

Real-time detection and recognition.



ACCURATE

High accuracy with AI and OCR.



MANAGEABLE

Easy to update vehicle records.



1. CAPTURE IMAGE

The camera captures the vehicle image.



2. DETECT PLATE

YOLO model locates the license plate.



3. READ PLATE

OCR extracts the characters.



4. VERIFY DATABASE

The system checks if the plate is authorized.



5. GRANT ACCESS

If authorized, the gate opens automatically.



AUTHORIZED

Gate Open



UNAUTHORIZED

Access Denied
Alert Security



TECHNICAL APPROACH

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ABC 1234

1 CV TECHNIQUE

DETECTION, MULTI COMPONENT



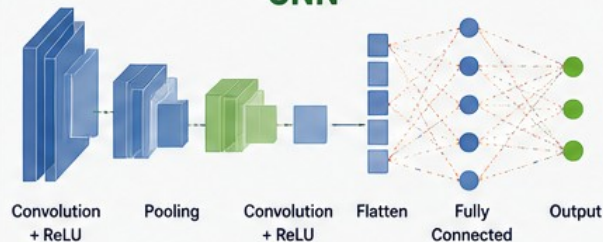
Computer Vision technique focused on detecting the license plate.



Multi component system that works with detection and OCR.

2 MODEL ARCHITECTURE

CNN



CNN is perfect for my project because the YOLO will be looking for the license plate that have distinguished shape, colors and have letters and numbers arranged in a line format.

3 MODEL

YOLO



YOLO Output

ABC 1234



Fast detection



High accuracy



Works very well with OCR



YOLO is the perfect model to detect the license plate, it is fast, high accurate and works very well with OCR.

4 HOW WE WILL USE IT

TRANSFER LEARNING



Pretrained YOLO Model



License Plate Dataset



Fine-tuned Model (Our Project)



My project will use the YOLO module that is already pretrained and I will tune it on a license plate database.

5 FRAMEWORK



YOLO is implemented and trained using PyTorch



Flexible and easy to use



Great performance and large community



Because YOLO is implemented and trained using PyTorch.

6 WHY THIS APPROACH?



SAFE

Reliable detection reduces false alarms and improves security.



FAST

YOLO provides real-time detection, ideal for gate access.



ACCURATE

High detection accuracy improves user satisfaction and trust.



This approach is **safe**, **fast** and **accurate**.

Use YOLO is perfect for plate detection and the transfer learning process will reduce the training effort.

PyTorch provides a reliable framework to develop the model.



OUR GOAL: Build a safe, fast, and accurate AI system that automatically detects and reads license plates to provide secure and seamless access for gated communities.





DATASET

License Plate Detection Dataset

- This dataset is used to train the AI model to detect license plates
- in a variety of real-world vehicles and conditions.



SOURCE

k kaggle



SIZE

10.145 images



LABELS

Detecting license plates



LINK

<https://www.kaggle.com/datasets/barkataliarbab/license-plate-detection-dataset-10125-images?resource=download>



This dataset contains **10.145 images** with annotated license plates, ideal for training YOLO models for accurate license plate detection.



DATASET OVERVIEW



10.145
Images



Various
Vehicle Types



Different
Lighting Conditions



Real-world
Environments



EXAMPLE IMAGES WITH ANNOTATIONS



High Quality
Clear and labeled
images



Diverse
Multiple countries,
plate styles and sizes



Real-World
Captured in different
conditions and angles



Perfect for AI
Ideal for training and
improving detection models



SUCCESS METRICS

Measuring the performance and impact of the AI License Plate Recognition System



PRIMARY METRIC: ACCURACY

I will measure accuracy and expect to achieve around **90% to 95%**.



TARGET ACCURACY



High accuracy is critical for a system that will be used by a large number of people.



If the system keeps triggering false alarms, residents can become frustrated and lose trust in the system.



GOAL:

Reliable detection and recognition with minimal false alarms to ensure resident satisfaction and system trust.



SECONDARY METRIC: SPEED & EFFICIENCY

Make the system work in the **fastest** and **most accurate** way.



FAST RESPONSE TIME

Detect and recognize license plates in real time to allow smooth and quick vehicle access.



HIGH THROUGHPUT

Handle a large number of vehicles efficiently without delays.



CONSISTENT PERFORMANCE

Maintain high accuracy even in different lighting, weather, and vehicle conditions.



WHY THESE METRICS MATTER



IMPROVE TRUST

High accuracy reduces false alarms and increases resident confidence.



SUPPORT SCALE

A reliable system can be used by a large number of residents and vehicles.



SAVE TIME

Fast and efficient processing creates a better experience for residents and security.



BETTER SECURITY

Accurate and fast recognition ensures only authorized vehicles enter the community.



TARGET: **90% – 95% ACCURACY** | FAST | RELIABLE | RESIDENT-FOCUSED

A smart system that delivers accuracy, speed, and security that residents can trust every day.





MILESTONE PLAN

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WEEK	PHASE	GOAL / ACTIVITIES	KEY DELIVERABLES
WEEK 1 (CURRENT) 	 BLUEPRINT	• Define the problem and objectives • Research and select dataset • Design system architecture • Plan tools and technologies	 Project plan, system diagram, dataset selected, and technology stack defined.
WEEK 2 	 FIRST WORKING DEMO	• Set up development environment • Load dataset and explore images • Run pre-trained YOLO model to detect license plates	 Working demo that detects license plates in sample images using YOLO.
WEEK 3 	 MAKE IT YOURS	• Fine tune YOLO model on the dataset • Integrate OCR (EasyOCR) to read license plate text • Display detected and read results	 Fine-tuned YOLO model and OCR successfully reading and detecting license plates.
WEEK 4 	 IMPROVE AND MEASURE	• Measure accuracy and performance • Check if system works as expected • Adjust model and parameters if needed • Validate results	 Performance report (accuracy, precision, recall), validation results, and improvements.
WEEK 5 	 BUILD	• Build the complete application • Connect database of authorized vehicles • Create user interface and integrate all components • Test full system and ensure it works	 Complete working application ready for use and final presentation.



PROJECT GOAL: Deliver a fast, accurate, and reliable AI system that automatically detects, reads, and verifies license plates to provide secure and seamless access for residents.



SUCCESS TARGET:
90% – 95% ACCURACY



RISKS & COST

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RISKS & MITIGATION

RISK	RISK LEVEL	MITIGATION STRATEGY
 DIFFERENT PLATE DESIGNS License plates vary by country, state, font, size, and color.	 MEDIUM	 Train the model with a diverse dataset. Use images with different plate styles, fonts, sizes, and backgrounds to improve generalization.
 POOR LIGHTING Low light, shadows, or glare can reduce detection and recognition accuracy.	 HIGH	 Train the model with images captured with different light. Include daytime, nighttime, shadows, backlight, and indoor/outdoor conditions in the training data.



COST



100% FREE SOLUTION

This project uses free tools, platforms, and resources to build a complete AI License Plate Recognition System.

FREE TOOLS & SERVICES USED



Google Colab

Free GPU/CPU for training



Kaggle

Free dataset



YOLO (Ultralytics)

Open source model



EasyOCR

Open source OCR library



Python

Open source programming language



By identifying risks and using free tools, the project aims to deliver an **accurate, reliable,** and **cost-effective** solution for gated community access.

